GSOC 2026

ML4SCI

PROJECT

Super Resolution at CMS Detector

Proposal by

Bhagavat Pratim Das

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About Me

Student Information

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University Information

University:	Indian Institute of Technology (ISM) Dhanbad
Major:	Mining Machinery Engineering
Current year:	2nd
Expected Graduation date:	August 2028
Degree:	B.Tech

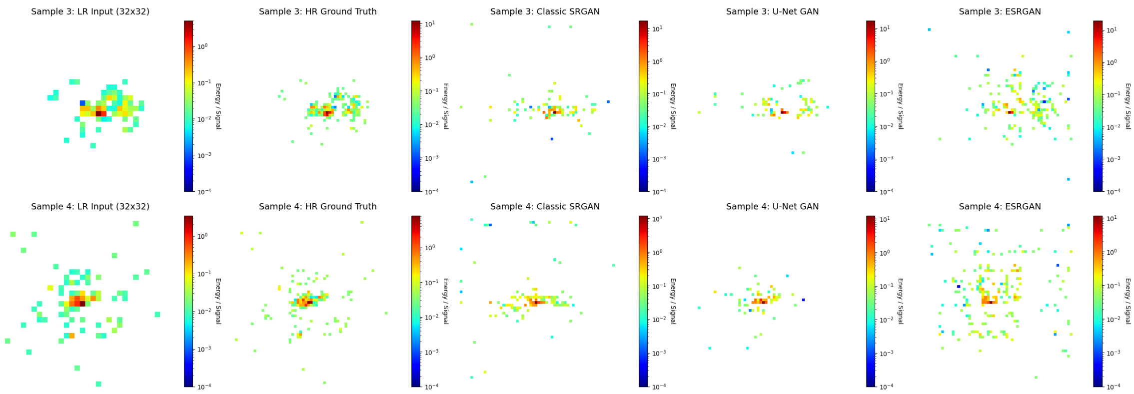
Background Work and Programming Skills

- I am a second-year undergraduate student at Indian Institute of Technology (ISM) Dhanbad, India. I am pursuing a degree in Mining Machinery Engineering.
- I use Google Colaboratory, Jupyter, and Kaggle notebooks for implementing Python code, and Visual Studio Code for development.
- I am proficient in C, C++ and Python, and I have a strong foundation in machine learning libraries including PyTorch, TensorFlow, and Scikit-learn. I am also comfortable with using Git and GitHub for version control.
- I possess a strong dedication to machine learning research, supported by solid expertise in research methodology and data analysis. Currently, I am exploring modern techniques that leverage transformer neural networks. Specifically, I am actively investigating Visual Autoregression, Joint Embedding Predictive Architectures (JEPA), and advanced Diffusion models to build highly effective predictive architectures.
- My interests lie in uncovering the underlying rules of complex physical systems through advanced machine learning algorithms. Currently, I am deeply exploring deep learning architectures, working extensively with PyTorch to build and debug complex models like Denoising U-Nets for conditional diffusion. I am particularly fascinated by how we embed inductive biases into these models to guide learning in meaningful ways, and ultimately, I aspire to develop architectures that can learn these physics-aware biases on their own.

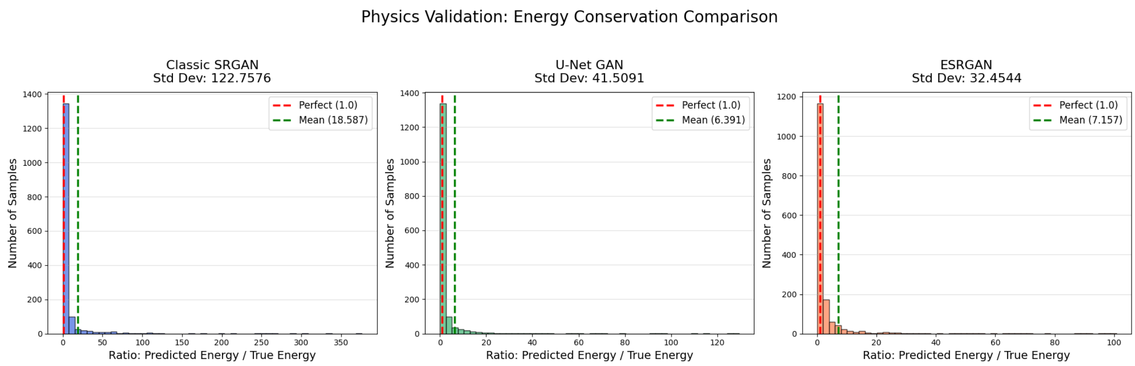
Evaluation Task

I have successfully completed the mandatory evaluation task for the Super-resolution at CMS detector project, adhering to the provided guidelines. I focused on upscaling low-resolution calorimeter images while strictly enforcing physical laws, structuring my project into dataset processing, multi-architecture GAN experimentation, and the development of custom physics-informed loss functions.

For dataset preparation, I processed the provided 125x125 three-channel matrices of low and high-resolution particle showers, formatting them to ensure stable adversarial training. Since the complete dataset file was too heavy to load entirely into the GPU memory at once, I implemented a data generator function to load the dataset streamingly in batches. The U-Net GAN was evaluated for its ability to maintain global spatial localization due to its skip connections, while the ESRGAN was evaluated for its capacity to reduce generation artifacts and improve overall structural quality. Comparing these distinct models allowed me to definitively understand the trade-offs between exact pixel reconstruction and perceptual shower quality.



A major focus of my optimization strategy was ensuring the networks respected fundamental physical laws, rather than just computer vision metrics. I optimized the training process by introducing custom physics-constrained loss functions. First, I implemented a custom metric designed specifically to penalize topology mismatching. This loss ensures that the generated high-resolution images accurately preserve the distinct physical scattering patterns and spatial profiles. Next, I introduced an L1 penalty that strictly enforces total energy conservation, penalizing the model whenever the sum of the generated high-resolution energies deviated from the low-resolution input. To strictly enforce positive energy, I used the ReLU activation function at the very end of my Generators.



The link to the GitHub repository containing the complete codebase, architectural implementations, and training scripts is [Link].

Initiative

Beyond the mandatory evaluation, I also took the initiative to build a custom diffusion model specifically to get a better understanding of architectural stability, identify potential training bottlenecks and validate physics informed loss integration with diffusion model before scaling to the full framework. I designed the network using Sparse ResNet layers combined with Window Attention blocks since the detector hits have a high degree of data sparsity. This structure allows the model to process the sparse spatial features efficiently without wasting computing power on empty areas. To prepare the data for the network, I preprocessed it using a log transformation to help stabilize the wide range of energy values.

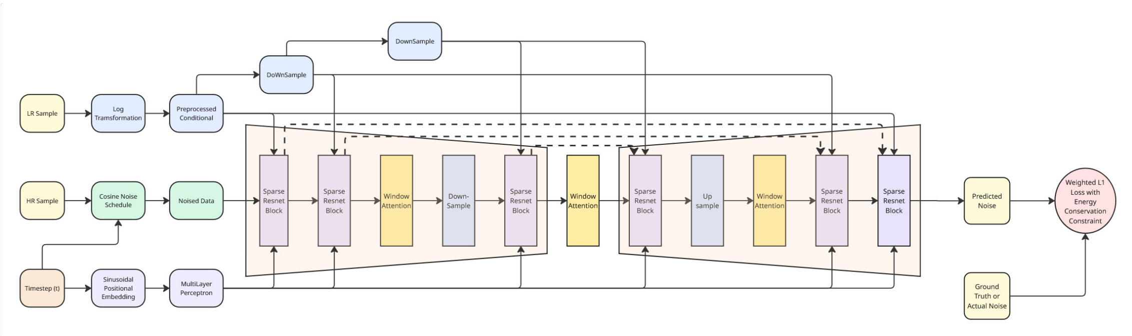
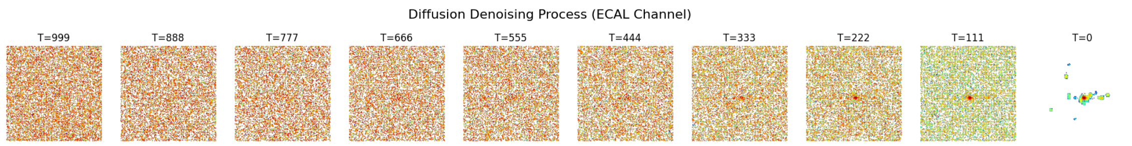
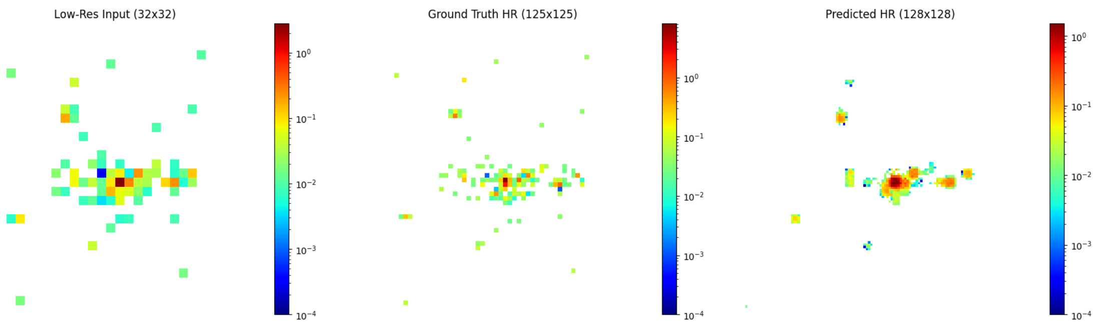


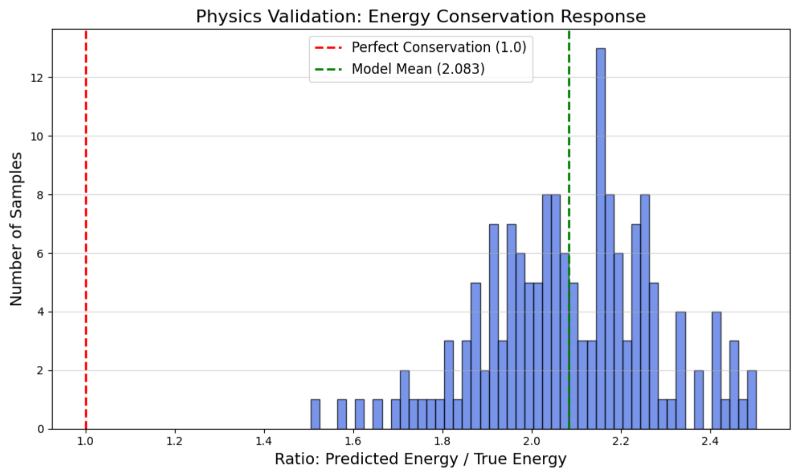
Figure 1: Diffusion Model Architecture

To train the model, I optimized it using a physics-constrained L1 loss. I specifically introduced penalties to ensure strict energy conservation, making sure that the total energy matched between the low and high-resolution data. Finally, positive energy constraints were applied to guarantee that the model only produces physically valid positive energy results.



Due to the constraints of free-tier GPU limits, I was only able to train this diffusion prototype for 50 epochs. However, the model demonstrated highly promising preliminary results — successfully mapping sparse inputs to high resolution and correctly localizing energy clusters. This strong proof-of-concept confirms the core architecture is sound and ready to scale with GSoC compute resources.





The link to the GitHub repository that contains the complete codebase is [Link].

Project Information

Organization: ML4SCI

Project: Super Resolution at The CMS Detector

Mentors:

- Eric Reinhardt (University of Alabama)
- Diptarko Choudhury (NISER)
- Ruchi Chudasama (University of Alabama)
- Emanuele Usai (University of Alabama)
- Sergei Gleyzer (University of Alabama)

Project Length: 175/350 hours

The energy and precision frontiers of high-energy physics are fundamentally driven by technological advances in our ability to record and reconstruct particle collisions. In the dense, pileup-heavy environments of the Large Hadron Collider (LHC), the imaging capabilities of calorimeters are paramount. High-granularity data is essential for Particle Flow algorithms to accurately map energy deposits to tracks and identify complex event topologies. However, achieving optimal spatial resolution is heavily constrained by immense financial costs and complex mechanical and electrical engineering challenges, often leading to less-than-ideal detector granularity in practice.

Machine learning-based super-resolution (SR) emerges as a transformative approach to bypass these hardware limitations. By reconstructing high-resolution (HR) data from low-resolution (LR) inputs, SR techniques computationally enhance calorimeter granularity without requiring physical modifications to the detector infrastructure. Unlike traditional interpolation methods, advanced generative algorithms such as Continuous Normalizing Flows (CNFs) and conditional diffusion models can learn the complex, underlying probability distributions of electromagnetic and hadronic showers.

This project aims to integrate state-of-the-art super-resolution techniques directly into the core LHC reconstruction pipeline. By employing continuous, geometry-independent representations like graphs or point clouds, the proposed framework will simultaneously suppress electronic noise and upsample sparse energy deposits.

Ultimately, this endeavor is about harnessing the power of advanced generative modeling to reveal fine-grained structural details within particle showers, offering a more precise window into the physical interactions captured by the CMS detector.

Abstract

- **Deep Learning & Inspiration:** Leverage advanced generative deep learning techniques specifically exploring transformer-based architectures such as Joint Embedding Predictive Architectures (JEPA) [Paper4], Visual Autoregression (VAR) [Paper3], and Latent Diffusion [Paper1], [Paper2], to accurately reconstruct high-resolution particle collision events from low-resolution calorimeter data. This draws inspiration from recent breakthroughs in self-supervised "world models" and continuous normalizing flows.
- **Model Development & Extension:** Develop and validate neural network models, establishing initial baselines with PyTorch implementations like Denoising U-Nets, to map processed, low-granularity simulated collisions back to a high-resolution representation. The aim is to progressively scale and combine these methodologies (e.g., utilizing JEPA as a physics encoder and a U-Net for generative decoding) to probe the highly sparse and complex phase space of CMS datasets.
- **Physics-Aware Benchmarking:** Build generative models that are strictly constrained by fundamental conservation laws and precise detector hardware limits. These models will be benchmarked against traditional baseline methods for superior data efficiency and downstream reconstruction accuracy, specifically evaluating physical metrics like energy correlator observables (C_2,C_3,D_2) and particle cardinality.

Ideas

I will be working this summer on developing, validating, and applying this novel hybrid framework:

1. **Develop Hybrid Super-Resolution Framework:** Implement and validate a deep learning pipeline combining a self-supervised physics encoder (JEPA) with a generative decoder (Diffusion and Visual Autoregression). Validation will occur on simulated calorimeter datasets to establish the model's ability to map low-granularity energy deposits to a high-resolution spatial representation.
2. **Enforce and Validate Physical Constraints:** Develop and test methods (e.g., custom loss functions, architectural masking, or discretization post-processing) to ensure the generative model strictly adheres to fundamental physical laws and hardware realities. This bridges the gap between pure AI generation and physical simulation via energy conservation, transverse momentum (p_T) balance, and detector-specific limits like ADC quantization and structural dead zones.

3. **Apply Super-Resolution to CMS Calorimeter Data:** Preprocess simulated CMS collision data (Likely using PARQUET formate, generated via GEANT4/COCOA) and apply the validated hybrid generative model to upsample these coarse detector readings, effectively acting as a High-Granularity Calorimeter (HGCAL) upgrade.

4. **Probe Shower Topology and Substructure:** Utilize the high-resolution outputs to analyze the shower topologies. Visualize and calculate critical jet substructure observables, specifically energy correlation functions like C_2, C_3 and D_2 to verify that the generated high-resolution output accurately differentiates true physics processes (e.g., hadronic vs. electromagnetic showers) without inventing fake patterns.

5. **Integrate with Particle Flow (PFlow) Reconstruction:** Utilize the generated high-resolution data to feed into a **Particle Flow (PFlow)** algorithm. This transformer based tool will use the generated details to count the exact number of particles (cardinality) and calculate their exact trajectories and energy (kinematics).

6. **Evaluate Performance and Data Efficiency:** Rigorously benchmark the super-resolution-enhanced PFlow pipeline against standard low-resolution baselines, focusing on improvements in the particle cardinality prediction accuracy, kinematic residuals (Δp_T, $\Delta \eta$, $\Delta \phi$), noise suppression capabilities and overall data efficiency when training the downstream reconstruction tasks.

In my view, the relevant skills required would be Python, PyTorch, Deep Learning (especially **Diffusion, Visual Autoregression models** and **Self-Supervised Learning**), and knowledge of **Collider Kinematics, Calorimetry** and **Jet Substructure**.

Approach

1. Data Pre-processing & Transformation

Motivation

Calorimeter data is difficult to process because it is extremely sparse and has a massive dynamic range. Raw data training is numerically unstable, and standard 2D grids waste computation on empty space while distorting the detector's irregular 3D geometry.

Methodology

- **Logarithmic Scaling:** Energy values (E) are compressed using $E' = log(E + \epsilon)$ to stabilize the neural network's gradients.
- **Graph Mapping:** Data is represented as sparse graphs where nodes are active detector cells. This avoids processing empty pixels and preserves the true physical layout of the CMS calorimeter.

Implementation

- **Point Cloud Construction:** Convert PARQUET files into graph structures using PyTorch Geometric (PyG), storing energy and (η, ϕ, r) coordinates as node features.
- Implement the Logarithmic Transformation after graph conversion. Apply $E = 10^{E'} - 1$ operation on high-resolution output to recover the original physical energy units.

2. Developing the Hybrid Super-Resolution Framework

Motivation

Existing methods for discovering continuous symmetries in datasets face three key limitations:

1. **Hardware Blindness:** They ignore physical detector limits (e.g., structural dead zones, analog-to-digital digitization ceilings), hallucinating impossible energy deposits in empty space.
2. **Kinematic Inconsistency:** They fail to mathematically enforce fundamental laws, such as local energy conservation and transverse momentum (p_T) balance.

3. **Computational Inefficiency:** They scale poorly when directly generating sparse, high-dimensional 3D/4D detector data.

This work proposes a novel hybrid framework that:

1. **Learns robust global physics dynamics** efficiently without relying on rigid, pixel-level target matching by leveraging self-supervised latent embeddings.
2. **Ensures the generated high-resolution transformations** are strictly bounded by fundamental conservation laws and hardware constraints, enabling scientifically reliable downstream Particle Flow reconstruction.

Methodology

1. Hardware Masking

- Parameterize physical hardware limits, such as structural dead zones and module boundaries as explicit binary masks M to mathematically restrict the generative space.

2. Physics-Aware Latent Encoding

- Train a self-supervised JEPA to capture global event topologies, shower shapes, and invariant mass relationships.
- Pass the low-resolution input through an encoder network E_θ to extract a dense, physics-aware latent embedding: $z_c = E_\theta(x_{LR})$.
- The self-supervised embedding loss is defined by:

$$L_{JEPA} = \|Predictor(z_c) - z_t\|_2^2$$

(where z_t is the latent representation of masked detector regions)

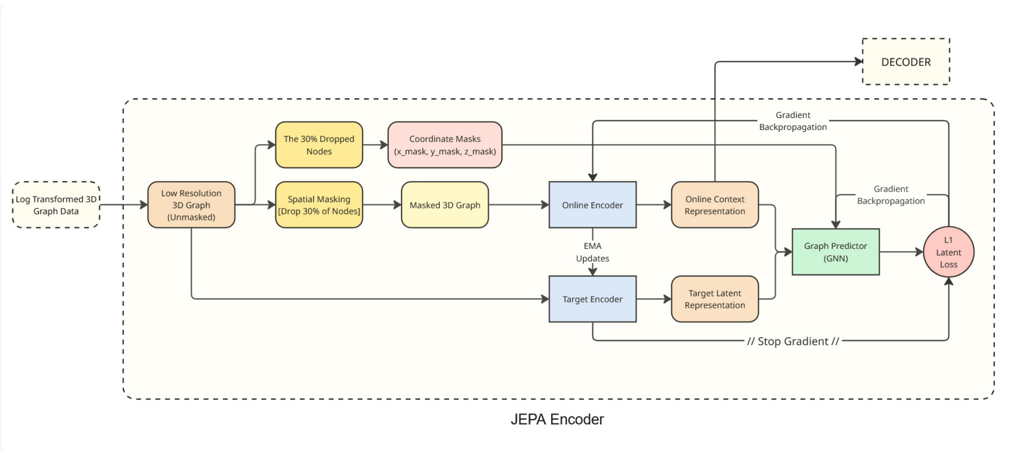


Figure 2: Physics-Aware JEPA Encoder

3. **Constrained Generative Decoding**

   - Implement a Denoising U-Net, an Auto-regression (optional) and a Continuous Normalizing Flow to generate the high-resolution spatial data.
   - Train both networks to reverse a forward noise process, predicting the high-resolution state $\hat{x}_{HR}$ at each step.
   - Enforce strict physical laws using a hybrid loss function:

   $$L_{Total} = L_{Diffusion}(\hat{x}_{HR}, x_{HR}) + \lambda \cdot Penalty(Energy(\hat{x}_{HR}) - Energy(x_{LR}))$$

     (where λ is the weighting factor that controls the balance between spatial pattern of data and physical constraints)
   - Select the Decoder with best performance.

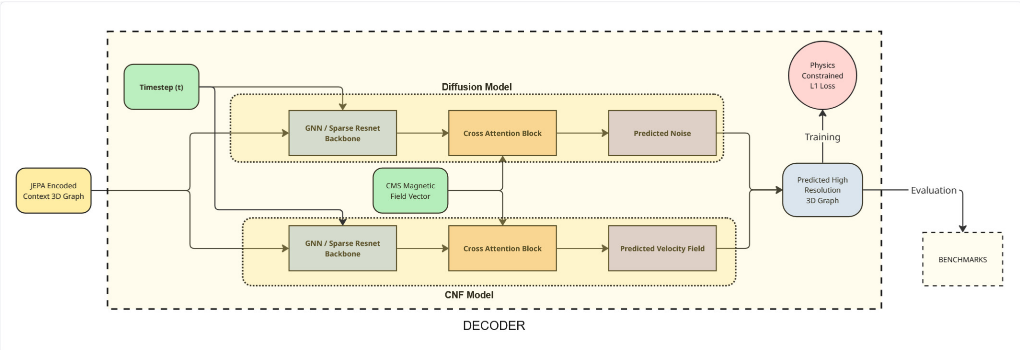


Figure 3: Physics Constrained Decoder

4. **Ensemble Sampling and Discretization**

   - During inference, execute multiple stochastic passes of the generative model for the exact same collision event to suppress artificial hallucinations.
   - Average these outputs to cancel out random generative fluctuations, yielding a stable, deterministic physical energy distribution:

   $$\hat{x}_{final} = \frac{1}{N} \sum_{i=1}^{N} \hat{x}_{HR}^{(i)}$$

Theoretical Justification

1. **Distributional Invariance:** The generative model acts as a flexible mapping function. By enforcing $p(x_{HR}) \approx p(\hat{x}_{HR})$, it accurately preserves complex shower topologies.
2. **Kinematic Consistency:** Explicit physics-based loss functions guarantee obedience to fundamental laws, such as local energy conservation and p_T balance.

3. **Physics-Informed Regularization:** Penalizing physically impossible predictions acts as a strict mathematical regularizer, stabilizing the training loop and prevents hallucinations.

3. Physics-Constrained Optimization and Inference

Motivation

Standard models optimize purely for visual similarity, remaining hardware blind and hallucinating impossible energy deposits. While the JEPA encoder successfully captures macroscopic physics like transverse momentum (p_T) balance and invariant mass, it cannot enforce pixel-level geometric rules such as local energy conservation, extreme sparsity, structural dead zones, magnetic asymmetries, and hardware quantization.

Methodology

1. **Local Energy Conservation:** Enforced by integrating a mean squared error (MSE) penalty term directly into the backpropagation cycle that penalizes gradient updates if total energy in generated output deviates from the low-resolution input.
2. **Strict Non-Negativity:** applying a ReLU activation function at the terminal layer of the decoder.This mathematically clips any negative continuous-space predictions to exactly zero.
3. **Dead Zones and Geometry:** Handled by multiplying the decoder's output tensors with a pre-calculated binary spatial mask that physically forces the energy gradients to 0 in coordinates overlapping the structural gaps between CMS modules.
4. **Magnetic Field Asymmetries:** Inject the CMS magnetic field vector as an explicit conditioning variable into the decoder's cross-attention layers, teaching the network to systematically bend the generated high-resolution pixels along the azimuthal axis(ϕ).
5. **Angular Boundaries:** Apply modulo arithmetic ($\phi \bmod 2\pi$) to the coordinate embeddings to set Angular Limits.

Implementation

Implement a custome loss function that penalizes energy deviations. Apply Binary spatial masks directly to the network's output layers to guarantee that exactly 0 energy is generated inside known dead zones.

4. Particle Flow Integration & Benchmarking

Motivation

Standard computer vision metrics (like MSE or SSIM) are entirely inadequate for evaluating high-energy physics data. To prove that this hybrid generative model acts as a valid, super-resolution model, its outputs must be rigorously evaluated on their ability to improve the downstream physical reconstruction of final-state particles.

Methodology

The super-resolved, physics-constrained energy graphs are evaluated by feeding them directly into a downstream Particle Flow (PFlow) algorithm. The pipeline compares the particles reconstructed from the generated high-resolution (HR) data against those reconstructed from baseline low-resolution (LR) data and ground-truth HR simulations.

Implementation

1. **Downstream MLPF Integration:** Ingest the generated sparse graphs by an MLPF algorithm (e.g., a GNN or Transformer) to predict the momentum, energy, and Particle ID (PID) of all stable particles.

2. **Reproducible Kinematic Benchmarks:** Inplement a benchmarking script that will compute the standard deviation of fractional transverse momentum resolution ($\Delta p_T / p_T$) and angular residuals ($\Delta\eta, \Delta\phi$) to quantify precision gains over the LR baseline.

3. **Topological Evaluation:** Implement a script for Jet Clustering via Anti-k_T algorithm, calculation of correlation functions (C_2, C_3) and their ratio (D_2) and plotting a statistical comparison (histogram plots) between the generated data and the ground truth data.

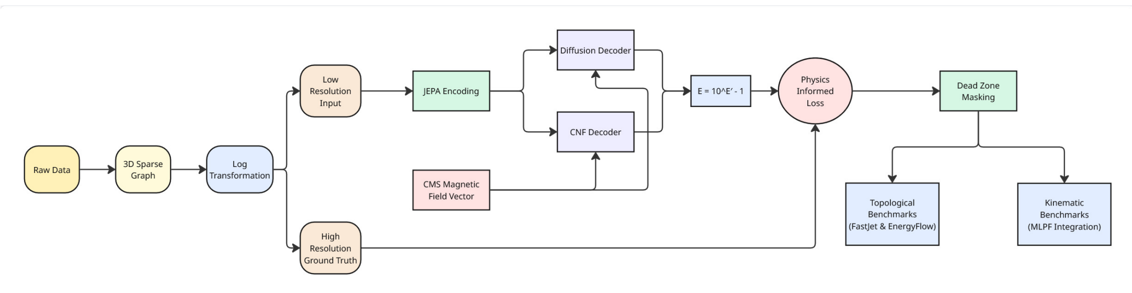


Figure 4: Super Resolution Framework

Timeline (Tentative)

Total Project Length: ~175 hours

From my past experiences, I can say that things can change in a minimal time. For example, some bug comes up with a high priority level that needs to be fixed before any more work. However, having a timeline is always suitable for any project. This, at least, helps in keeping a tab on how much you are lagging.

Pre-GSOC Period

Before diving into coding at the start of June, there's extensive groundwork for me to cover.

- Thoroughly study the key research papers: Denoising Graph Super-Resolution, Conditional Diffusion Model for Super-Resolution (DiffLense), Visual Autoregressive Modeling, V-JEPA 2.
- Deepen understanding of relevant concepts: Joint Embedding Predictive Architectures, Autoregression and Diffusion, Collider Kinematics, Jet Substructure and Particle Flow algorithm.

I am planning to split the project in four phases:

- Data Engineering & JEPA Pre-training
- Generative Decoder Architectures & Physics Constraints Implementation
- Hardware Alignment & Scaled Training
- PFlow Integration, Benchmarking & Wrap-up

Timeline

Period	Tasks
Phase 1: Data Engineering & JEPA Pre-training	
Week 1 [May 25 - May 31]	• Finalize project structure (Git repo, modules). • Set up data loading pipelines for low-resolution and high-resolution particle collision data. • Implement filtering logic and map the active hits into 3D sparse graphs using PyTorch Geometric. • Apply log transformation $E' = log_{10}(E + 1)$
Week 2 [June 1 - June 7]	• Construct the Joint-Embedding Predictive Architecture (JEPA) using Graph Neural Network (GNN) layers. • Implement the spatial masking strategy to hide sections of the LR detector graphs.
Week 3 [June 8 - June 14]	• Run the self-supervised training loop. • Implement the Exponential Moving Average (EMA) updates for the target encoder.
Phase 2: Generative Decoders & Physics Constraints Implementation	
Week 4 [June 15 - June 21]	• Implement Denoising Diffusion Probabilistic Model (DDPM) backbone adapted for 3D graph structures. • Implement the Cross-Attention mechanism to inject the CMS magnetic field vector as conditioning.
Week 5 [June 22 - June 28]	• Implement the Continuous Normalizing Flow architecture, sharing the GNN backbone.
Week 6 [June 29 - July 5]	• Inject mathematical physics constraints into the loss functions. • Implement the MSE penalty term: $L_{Total} = L_{Gen} + \lambda \cdot (\sum E_{LR} - \sum E_{HR})^2$ for energy conservation, and apply terminal ReLU activations for non-negativity ($E \geq 0$).
MID EVALUATION [July 6 to July 12]	
Week 7 [July 13 - July 19]	• Train the pipeline on a simplified dummy subset of the data to verify convergence. • Systematically tune the hyperparameter λ to balance generation and physics preservation.
Phase 3: Hardware Alignment & Scaled Training	

Period	Tasks
Week 8 [July 20 - July 26]	• Integrate binary masks to explicitly force energy gradients to 0 inside known CMS structural dead zones. • Implement ensemble sampling inference to produce deterministic, noise-reduced measurements.
Week 9 [July 27 - August 2]	• Migrate both stabilized models from the dummy dataset to the full, complex CMS simulated dataset. • Implement gradient accumulation and mixed-precision training to handle massive sparse graphs without Out-Of-Memory errors.
Phase 4: PFlow Integration, Benchmarking & Wrap-up	
Week 10 [August 3 - August 9]	• Feed the HR outputs into a downstream Particle Flow algorithm. • Calculate and plot the fractional transverse momentum resolution ($\Delta p_T/p_T$) and angular residuals ($\Delta\eta, \Delta\phi$).
Week 11 [August 10 - August 16]	• Integrate FastJet and EnergyFlow (Tools) to cluster the generated hits and use EnergyFlow to calculate Energy Correlators (C_2, C_3, D_2). • Run statistical tests to compare Diffusion vs. CNF on preserving the physical clumpiness of jets.
Week 12 [August 17 - August 23]	• Finalize all analysis, address any critical blockers or high-priority bugs. • Clean the repository, finalize documentation ensuring the benchmark scripts are fully reproducible.
FINAL EVALUATION [August 24 to August 30]	

By the end of the last week (week 12), I will ensure that all of the above implementations have been documented and tested. I will extend this period to my Future Work by writing blogs explaining all the works and if my work leads to a research paper, I will want to be a part of it.

I will update my progress weekly to my mentors and incorporate their feedback and suggestions as and when required. I am confident enough to complete the project on time. I have also kept a buffer week before final evaluations for any critical blockers.

Deliverables

- A framework capable of processing CMS datasets into 3D sparse graphs and executing physics-constrained generative super-resolution.
- Physics-aware generative models (Diffusion and Continuous Normalizing Flows) incorporated with local energy conservation and geometric dead-zone masking.
- Benchmark evaluations quantifying the model's physical accuracy in downstream tasks, specifically measuring kinematic resolutions and topological jet substructure.
- A GitHub repository containing all of the above implementations, including the data extraction pipeline, model architectures, and training scripts.
- Beginner-friendly README for easy installation, detailing exact environment setup and steps for reproducible training and benchmarking.

Future Scope

- Deploying the optimized generative model for ultra-low latency inference in the CMS High-Level Trigger (HLT).
- Synthesizing the decoder onto FPGAs for real-time edge computing at the detector level.
- Expanding the JEPA graph encoder to accept heterogeneous inputs.

The Motivation for GSOC

I am a dedicated programmer and problem-solver, deeply passionate about machine learning and a member of Animation and Game Designing Club. I find joy in understanding algorithms that allow models to learn patterns and evolve. I also have a keen interest in mathematics and physics, specifically how machine learning connects with traditional sciences. Contributing to open-source projects is a fulfilling way for me to collaborate with diverse communities, improving both my technical and teamwork skills. The prospect of creating solutions that could positively impact millions of lives fills me with immense pride, and I'm constantly motivated to expand my knowledge by delving into research papers and staying updated with the latest advancements in technology.

Achievements

- Qualified JEE Mains and JEE Advanced
- Codeforces: Pupil (Maximum Rating - 1203)
- First Runner-Up: Aagglomeration 2.0 Hackathon, IIT(ISM) Dhanbad (2026)
- Participant: Hackxios hackathon, IIIT Bhopal (2025)
- Participant: AlgoUtsav competitive programming competition (2026)

∼ THANK YOU ∼